

Curriculum Vitae

Personal information



Name **Qazi Muhammad Saqib, PhD**

Assistant Professor, Department of Electrical and Electronic Engineering
Sivas University of Science & Technology (SBTÜ), Türkiye
Principle Research Scientist & Team Lead, Sustainability & Renewable Energy Research Laboratory, SBTÜ, Türkiye
Visiting Professor & Senior Advisor at OISE LAB, Jeju National University, Korea
Distinguished Brain Pool Alumnus (Prestigious Fellowship of South Korea)
Former Invited Research Scientist, Korea
Best Researcher of Korea nationwide Award of 2022
Permanent Resident of South Korea

Phone +90-501-5959270

E-mail gazisaqib12@gmail.com

Google Scholar [Google Scholar Profile](#)

ResearchGate [ResearchGate Profile](#)

ORCID [ORCID Profile](#)

Major and Specializations

Energy & Sustainability, Self-Powered Energy Generation & Storage Systems, Energy from Environment, Flexible Electronics, Electronic-Skin, Soft-Robotics, Self-powered Sensing

Teaching Areas and Academic Expertise

Basic and Advanced Electronics, Power Electronics, Circuit Analysis, Electrical Circuits, Electronics I & II, Analog Electronics, Digital Electronics, Electronic Devices, Semiconductor Devices, Instrumentation & Measurement, Microprocessors, Renewable Energy Systems, Electrical & Electronic Engineering Laboratory. Flexible Electronics, Nanotechnology, Organic Electronics, Optoelectronics, Semiconductor Engineering, Sensors & Systems & related subjects.

Publications in Top 1–5% High-Impact Journals (Last Three Years) (Cumulative IF > 600)

Publications in top 1% high-impact journals

- *eScience* (IF 50+)
- *Advanced Materials* (IF 30.85)
- *Materials Science and Engineering: R* (IF 26.8)

Publications in top 2% high-impact journals

- *International Journal of Extreme Manufacturing* (IF 24)
- *Advanced Composites and Hybrid Materials* (IF 23.1)

Publications in top 3% high-impact journals

- *Advanced Functional Materials* (IF 18.86, **3 articles**)
- *Nano Energy* (IF 17.88, **7 articles**)

Publications in top 4% high-impact journals

- *Advanced Science* (IF 15.1)
- *Chemical Engineering Journal* (IF 13.3, **3 articles**)
- *Nano Today* (IF 13.2)
- *Journal of Materials Chemistry A* (IF 12.73)

Publications in top 5% high-impact journals

CV Highlights

- *Small* (IF 12.1)
- *Materials Horizons* (IF 10.7)
- *ACS Applied Materials & Interfaces* (IF 9.0)
- *Journal of Energy Storage* (IF 8.9)
- *Sensors and Actuators B: Chemical* (IF 8.0, **3 articles**)
- *Journal of Power Sources* (IF 8.0)

Professional Experience

- Post-PhD Research Experience > 4.5 years,
- **Total Teaching Experience > 3.5 years**

Research Accomplishments

- **Funded Research Projects:** 6 (PL: 2, Participating Researcher: 4)
- **Highly competitive Brain Pool fellowship Alumnus (success rate 10-15%, 2022):** Recognized as a **Distinguished Brain Pool Alumnus** by the National Research Foundation (NRF) of Korea
- **Citations:** 1670 + Google Scholar Citations (i10-index 40+)
- **Cumulative Impact Factor:** 600+
- **Peer Reviewed Journal Publications:** 60+
- **Patent:** 1
- **Number of Published Book Chapters:** 3
- **Student Mentoring:** PhD: 4 MS: 1, Undergraduate: 4

International Collaboration and Recognition

- **Research collaboration:** Singapore, Australia, South Korea, UAE, USA, UK, Qatar, China, Saudi Arabia, & Spain, Turkey
- **International Research/Academic Event Participation:** Spain, Japan, South Korea, and Pakistan

Work experience

Dates	July 2025 ~ Present
Job Title	Principle Research Scientist, Sustainability & Renewable Energy Research Lab
Name and Address of Employer	Assistant Professor, Electrical Eng., Sivas University of Science & Technology, Türkiye
Main Activities and Responsibilities	- Principle Research Scientist-Sustainability and Renewable Energy Research Laboratory - Undergraduate Teaching and mentorship - Graduate Teaching and mentorship - Course material preparation, Lecture delivering, conducting exams, and students' evaluation. - Actively involved in various academic communities, and scientific meetings and international collaborations - Involved in extra departmental responsibilities - Carried out research and published multiple top-category Journal publications - Responsible for multiple successfully funded grants for Sustainable energy projects. - Administrative services and training of the faculty member and researchers - Established National and International collaborations

Courses Teaching	- <i>Electronics I (Theory & Laboratory)</i> - <i>Advanced Electronics (Theory & Laboratory)</i> - <i>Power Electronics (Theory & Laboratory)</i> - <i>Industrial Electronics (Theory & Laboratory)</i> - <i>Circuit Theory (Theory & Laboratory)</i> - <i>Optoelectronics (Theory & Laboratory)</i> - <i>Nanotechnology & Biosensors</i>
Dates	Jan 2025 ~ to date
Name and Address of Employer	Visiting Professor, Advisor OISE Laboratory, Jeju National University (JNU), Jeju, Korea
Main Activities and Responsibilities	- Visting Lectures and Demonstrations - Mentoring Graduate students and providing guidance on experimental design etc. - Strengthening research partnerships between institutions through collaborative projects.
Dates	July 2022 ~ Dec 2024
Job Title	Invited Research Scientist by Korean Government / Brain Pool Fellow / Advisor
Name and Address of Employer	OISE LAB, Jeju National University (JNU), Jeju, Korea
Main Activities and Responsibilities	- Worked under Brain Pool (BP) program awarded by the Ministry of ICT & NRF, South Korea - Carried out research and published multiple top-category Journal publications - Responsible for multiple successfully funded grants from NRF, Korea for Sustainable energy generation/ Electrochemical Storage Systems, and Sea water-based hydrogen production - Project Lead (PL) "Micro-scale Self-powered Energy Storage Systems for Energy Harvesting" - Co-Advisor of PhD, MS, and undergraduate students - Synthesizing advanced composite & hybrid materials for energy storage and OER/HER - Administrative services and training of the faculty member and researchers - Established National and International collaborations
Dates	March 2023 ~ Dec 2024
Job Title	Lecturer (Teaching)-Jeju National University (JNU), Jeju, Korea (MS and Doctoral Students)
Name and address of Employer	Materials and Electronic System Engineering Track, Ocean System Engineering, Jeju National University (JNU), Korea
Research Activities and Responsibilities	Courses Taught: - <i>Advanced Electronics</i> - <i>Basic Electronics and Circuit Theory</i> - <i>Sensors and Systems</i> - <i>Optoelectronics</i> - <i>Organic Electronics</i> - <i>Printed Electronics and Wearable Technologies</i> - <i>Nano-Engineering</i> - <i>Basic Electronics and Photo Engineering</i> - <i>Nanotechnology</i>
Dates	Feb. 2019 ~ Dec. 2019
Job Title	Display Scientist/Device Engineer
Name and address of Employer	Kyunghee University, Seoul, Korea
Research Activities	Managed Multiple research projects on Photovoltaics Research activities: Organic Light Emitting diodes (OLEDs), Perovskite based LEDs, Solution processed OLEDs, Inkjet Printed OLEDs
Education	
Date of Graduation	18 February 2022

Title of qualification awarded	Doctor of Philosophy (PhD) in Material and Electronic System Engineering (2019-2022)
Name of Institute	Jeju National University, Jeju-si, Jeju-do, Korea
Thesis Title	Self-powered Nanogenerators and Active Sensors Based on Advanced Functional Materials <i>This thesis is primarily concerned with bio-mechanical triboelectric and piezoelectric energy harvesting, micro-scale particles movement towards energy generation, self-powering behavior of triboelectric and piezoelectric devices, and capacitive pressure sensors. Furthermore, various advanced functional materials were explored for low-power energy harvesting and active sensing.</i>
Principal subjects	Organic Device Engineering, Energy System Engineering, Smart Materials and Applications, Advanced Polymer Engineering, Semiconductor Devices, Photo Device Engineering
Date of Graduation	23 February 2018
Title of qualification awarded	Master of Science in Electronics Engineering (2016-2018)
Name of Institute	Kyungpook National University, Daegu, South Korea
Thesis Title	Self-tuning smart window based on cholesteric liquid crystal for energy-saving building <i>This thesis investigates cholesteric liquid crystal-based smart windows featuring three distinct operational modes. Furthermore, the fabricated smart window shows self-tuning behavior across varying outside environments for energy-saving building.</i>
Principal subjects	Electronic Properties of Solid-State Devices, Display Device Engineering, Electronic Materials Engineering, Introduction to Display Engineering, Thin Film Engineering, Fundamental of Modern Optics
Date of Graduation	05 May 2012
Title of qualification awarded	Bachelor of Engineering in Electronic Engineering (2008-2012)
Name of Institute	Dawood University of Engineering and Technology, Karachi, Pakistan
Principal subjects	Electrical Circuit, Electronics, Communication Networks, Digital Logic Design, Industrial Electronics, Automation and Robotics, Control Engineering

Selected Publications

Peer-Reviewed Journal Articles

1. **Q. M. Saqib**, et al., Fe/Fe₃C Embedded Pyrazine-Based Porous Organic Network for Superior Tribo-charge and Electrochemical Double Layer Capacitance, [Advanced Functional Materials](#) (2025) (IF: 18.86).
2. **Saqib, Q.M.**, Ahmad, I., Mannan, A., Mahmood, J., Ameen, S., Patil, C.S., Noman, M., Kim, J., Okyay, M.S., Patil, S.R. and Ko, Y., **2024**. Triboelectric Energy Harvesting from Highly Conjugated Fused Aromatic Ladder Structure Under Extreme Environmental Conditions. [Advanced Materials](#), p.2311029 (IF: 29.4).
3. M. Y. Chougale, S. B. Ghode, S. L. Patil, J. Kim, **Saqib, Q.M.**, et al., Humidity-Responsive Multistate Resistive Switching Device for Non-Contact Sensory Neuro-Electronic Applications, [eScience](#), 100520 (2025) (IF: 36.6).
4. Baig, **Saqib, Q.M. (Joint First Author)**, M., Noman, M., Sheeraz, M., Rasheed, A., Yousuf, M., Lee, E., Kim, J., Ko, Y., Patil, C.S. and Patil, S.R., **2024**. Novel Intercalation Approach in MXene Using Modified Silica Nanospheres to Enhance the Surface Charge Density for Superior Triboelectric Performance, [Advanced Functional Materials](#), p.2408271 (IF: 18.86).
5. Noman, M., Baig, M.M., **Saqib, Q.M. (Joint First Author)**, Patil, S.R., Patil, C.S., Kim, J., Ko, Y., Lee, E., Hwang, J., Lee, S.G. and Bae, J., 2024. Ti₃C₂Tx-MXene based 2D/3D Ti₃C₂-TiO₂-CuTiO₃ heterostructure for enhanced pseudocapacitive performance. [Chemical Engineering Journal](#), p.156697 (IF: 13.3)

6. **Q. M. Saqib**, et al., 3-Dimensional Multistate Memristor Structures Based Neuromorphic Devices for High-Density In-Memory Computing, *Materials Science and Engineering: R* (Accepted, 2025) (IF: 26.6)
7. Q. M. Saqib, et al., Micro-space triboelectric nanogenerator based on Biocompatible lignin-derived activated carbon for advanced active sensing applications *ACS Applied Materials & Interfaces Journal* (Accepted, 2025) (IF: 9)
8. CS Patil, J Kim, SB Ghode, Paranjape, Noman, **QM Saqib, et al.**, High Flexible, Ultra-Robust Piezoelectric Nanogenerator Based on Hexagonal α -Beryllium Oxide/PVDF Composite Thin Film: E-Skin and Energy Harvesting Applications, *Advanced Functional Materials* (Accepted, 2025) (IF: 18.86).
9. SR Patil, R Kulkarni, SB Ghode, **QM Saqib, et al.**, Unraveling the Electrocatalytic Properties of Non-Noble Metal-Based Nickel-Cobalt-Tungsten/MXene Heterocomposite Towards Trifunctional Hydrogen Evolution Reaction, Oxygen, *Journal of Power Sources*, 652, 237675 (2025) (IF: 9.13).
10. **Q. M. Saqib, et al.**, Advances in Tribonegative Layer for 2D Material-Based Triboelectric Nanogenerators: Material Modifications and Architectural Designs, *Chemical Engineering Journal*, 163854 (2025) (IF: 15.1).
11. SB Ghode, CS Patil, J Kim, VP Bhandigare, J Kim, A Mannan, **Q. M. Saqib, et al.**, Plasma-Assisted Surface Modification and Formation of Oxygen Vacancies in Coffee-Derived Carbon for Triboelectric Energy Harvesting and Electrocatalytic Water Splitting, *Nano Energy*, 111116 (2025) (IF: 17.6).
12. SR Patil, R Kulkarni, SB Ghode, J Kim, **QM Saqib et al.**, Plasma-Assisted Surface Engineering of Binary Metal Chalcogenides: A Path Toward High Energy Efficiency, Electrocatalysts for Water Splitting, and Urea Oxidation with Stability ..., *ACS Applied Energy Materials*, 8 (4), 2346–2359 (2025) (IF: 6.0)
13. Patil, C.S., **Saqib, Q.M.**, Kim, J., Noman, M., Patil, S.R., Ko, Y., Kurkuri, M.D. and Bae, J., **2024**. Synergistic integration of graphene quantum dots into metal–organic framework-5 for enhancing triboelectric nanogenerator performance. *Advanced Composites and Hybrid Materials*, 7(5), pp.1-22 (IF: 23.1).
14. **Saqib, Q.M.**, Mannan, A., Noman, M., Chougale, M.Y., Patil, C.S., Ko, Y., Kim, J., Patil, S.R., Yousuf, M., Shaukat, R.A., Pyo Jeon, Y. and Bae, J. **2024**. Miniaturizing Power: Harnessing Micro-Supercapacitors for advanced micro-electronics. *Chemical Engineering Journal*, 490 (IF: 13.3)
15. Noman, M., **Saqib, Q.M. (Joint First Author)**, Ameen, S., Patil, S.R., Patil, C.S., Kim, J., Ko, Y., Kim, B. and Bae, J., **2024**. Controlling Triboelectric Charge of MOFs by Leveraging Ligands Chemistry. *Advanced Science*, p.2404993 (IF: 15.1).
16. **Saqib, Q.M.**, Yousuf, M., Noman, M., Mannan, A., Patil, C.S., Kim, J., Patil, S.R., Ko, Y., Chodankar, N.R. and Bae, J., **2024**. Arbitrary Directional Triboelectric Nanogenerators: Advanced Energy Harvesting for Sustainable Future. *Nano Energy*, p.110456., (IF: 17.6). **(Featured as Back Cover Article)**
17. Patil, C.S., **Saqib, Q.M.**, Patil, S.R., Noman, M., Chougale, M.Y., Shaukat, R.A., Kim, J., Ko, Y., Dongale, T.D. and Bae, J., **2024**. Triboelectric nanogenerator based on reactivated electrode materials derived from waste alkaline battery: Influence of pyrolysis temperature and surface morphology. *Nano Energy*, 121, p.109205 (IF: 17.6).
18. Patil, S.R., Kulkarni, R., Talib, S.H., Ghode, S.B., **Saqib, Q.M.**, Noman, M., Patil, C.S., Kim, J., Ko, Y., Dongale, T.D. and Al Ghaferi, A., 2025. Unraveling the electrocatalytic properties of non-noble metal-based nickel-cobalt-tungsten/MXene heterocomposite

- towards trifunctional hydrogen evolution reaction, oxygen evolution reaction, and methanol oxidation reaction. *Journal of Power Sources*, 652, p.237675. (IF: 8).
19. Khan, S.A., **Saqib, Q.M.**, Bae, J. and Cho, S.W., 2025. Advances in tribonegative layer for 2D material-based triboelectric nanogenerators: Material modifications and architectural designs. *Chemical Engineering Journal*, p.163854. (IF: 13.3)
 20. Ghode, S.B., Patil, Kim, J., Bhandigare, Mannan, A., Ko, Patil, Noman, M., **Saqib, Q.M.** and Chougale, M.Y., 2025. Plasma-assisted Surface Modification and Formation of Oxygen Vacancies in Coffee-derived Carbon for Triboelectric Energy Harvesting and Electrocatalytic Water Splitting. *Nano Energy*, p.111116. (IF: 17.6). (Featured as Back Cover Article)
 21. Patil, S.R., Kulkarni, R., Ghode, S.B., Kim, J., **Saqib, Q.M.**, Noman, M., Patil, C.S., Ko, Y., Bae, S.Y., Chang, Y.Y. and Koduru, J.R., 2025. Plasma-assisted surface engineering of binary metal chalcogenides: a path toward high energy efficiency, electrocatalysts for water splitting, and urea oxidation with stability prediction via machine learning. *ACS Applied Energy Materials*, 8(4), pp.2346-2359. (IF: 5.4).
 22. Patil, C.S., Noman, M., Ghode, S.B., Patil, S.R., Kim, J., **Saqib, Q.M.**, Samage, A., Kim, S.H., Jeon, Y.P. and Bae, J., 2025. Reactivated Cathode Material from Waste Alkaline Batteries for Supercapacitor Electrodes: Interplay Study Between Pyrolysis Temperature and Surface Morphology. *Small*, p.2502556. (IF: 12.1)
 23. Shaukat, R.A., Noman, M., **Saqib, Q.M. (Joint First Author)**, Umair, A., Baig, M.M., Yousuf, M., Chougale, M.Y., Kim, J., Patil, S.R., Patil, C.S. and Lee, S.G., 2024. Multi-layered quasi-2D perovskite based triboelectric nanogenerator. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, p.135728. (IF: 4.9)
 24. Patil, S.R., Noman, M., Kulkarni, R., Patil, C.S., **Saqib, Q.M.**, Chougale, M.Y., Kim, J., Ko, Y., Jeon, Y.P., Dongale, T.D. and Kang, J.C., 2024. Unveiling the marine Sargassum horneri material for energy and active sensor devices: Towards multitasking approaches. *Nano Today*, 57, p.102379 (IF: 13.2).
 25. Patil, S.R., Chougale, M.Y., Kim, J., Shaukat, R.A., Noman, **Saqib, Q.M.**, Q.M., Patil, C.S., Ghode, S.B., Dongale, T.D., Dubal, D. and Bae, J., 2024. Nature-Driven Edible Black Soldier Fly (BSF) Insect Larvae Derived Chitin Biofilm for Sustainable Multifunctional Energy Harvesting. *Advanced Sustainable Systems*, 8(2), p.2300312 (IF: 7.26)
 26. Noman, M., Shaukat, R.A., Patil, S.R., Chougale, M.Y., Kim, J., Patil, C.S., **Saqib, Q.M.**, Lee, H.W., Kim, M.S. and Bae, J., 2024. Triboelectric nanogenerator based on coastal bio-waste Ulva lactuca from Jeju island for sustainable energy harvesting. *Journal of Science: Advanced Materials and Devices*, 9(1), p.100669. (IF: 8.9)
 27. Kim, J., Chougale, M.Y., Shaukat, R.A., Patil, S.R., Noman, M., Patil, C.S., **Saqib, Q.M.**, Ko, Y. and Bae, J., 2024. F8BT-based highly sensitive humidity sensor for metaverse interfacing. *Sensors and Actuators B: Chemical*, 398, p.134717 (IF: 8.4).
 28. **Saqib, Q.M.**, Shaukat, R.A., Chougale, M.Y., Khan, M.U., Kim, J. and Bae, J., 2022. Particle triboelectric nanogenerator (P-TENG). *Nano Energy*, 100, p.107475 (IF: 17.6).
 29. Shaukat, R.A., **Saqib, Q.M. (Joint First Author)**, Kim, J., Song, H., Khan, M.U., Chougale, M.Y., Bae, J. and Choi, M.J., 2022. Ultra-robust tribo- and piezo-electric nanogenerator based on metal organic frameworks (MOF-5) with high environmental stability. *Nano Energy*, 96, p.107128 (IF: 17.6).
 30. **Saqib, Q.M.**, Chougale, M.Y., Khan, M.U., Shaukat, R.A., Kim, J., Bae, J., Lee, H.W., Park, J.I., Kim, M.S. and Lee, B.G., 2021. Natural seagrass tribopositive material based spray coatable triboelectric nanogenerator. *Nano Energy*, 89, p.106458 (IF: 17.6).

31. Noman, M., Shaukat, R.A., **Saqib, Q.M.**, Chougale, M.Y., Kim, J., Patil, S.R., Patil, C.S. and Bae, J., 2023. Sea urchin shaped hierarchical cobalt carbonate hydroxide hydrate-PANI nanocomposite with enhanced redox activity for asymmetric supercapacitor. *Journal of Energy Storage*, 73, p.109259 (IF: 8.9).
32. Shaukat, R.A., Ameen, S., **Saqib, Q.M.**, Chougale, M.Y., Kim, J., Patil, S.R., Noman, M., Kim, H.K. and Bae, J., 2023. Organic semiconductor polymers: a carbazole-based novel tribopositive polymer for energy harvesting with high temperature stability. *Journal of Materials Chemistry A*, 11(27), pp.14800-14808 (IF: 12.7).
33. **Saqib, Q.M.**, Chougale, M.Y., Khan, M.U., Shaukat, R.A., Kim, J., Bhat, K.S. and Bae, J., 2022. Triboelectric nanogenerator based on lignocellulosic waste fruit shell tribopositive material: Comparative analysis. *Materials Today Sustainability*, 18, p.100146 (IF: 8.57).
34. Chougale, M.Y., **Saqib, Q.M. (Joint First Author)**, Khan, M.U., Shaukat, R.A., Kim, J., Dubal, D. and Bae, J., 2021. Expired Pharmaceutical Drugs as Tribopositive Material for Triboelectric Nanogenerator (Adv. Sustainable Syst. 12/2021). *Advanced Sustainable Systems*, 5(12), p.2170023 (IF: 7.26). (Featured as Front Cover Article)
35. Khan, M.U., **Saqib, Q.M.**, Chougale, M.Y., Shaukat, R.A., Kim, J. and Bae, J., 2021. Soft and flexible: core-shell ionic liquid resistive memory for electronic synapses. *Microsystems & nanoengineering*, 7(1), p.78 (IF: 8.0).
36. Chougale, M.Y., Khan, M.U., Kim, J., Cosgrove, J., Shaukat, R.A., **Saqib, Q.M.**, Banjade, M., Patil, S.R., Brown, C., Dubal, D. and Bae, J., 2023. Snake ecdysis: A potential e-material for advanced electronic technology. *Nano Energy*, 111, p.108399 (IF: 17.6).
37. Shaukat, R.A., Khan, M.U., **Saqib, Q.M.**, Chougale, M.Y., Kim, J., Bermak, A. and Bae, J., 2022. Two dimensional Zirconium diselenide based humidity sensor for flexible electronics. *Sensors and Actuators B: Chemical*, 358, p.131507 (IF: 8.4).
38. Khan, M.U., Kim, J., Chougale, M.Y., Shaukat, R.A., **Saqib, Q.M.**, Patil, S.R., Mohammad, B. and Bae, J., 2023. Advancement in soft iontronic resistive memory devices and their application for neuromorphic computing. *Advanced Intelligent Systems*, 5(2), p.2200281 (IF: 7.4).
39. Patil, S.R., Chougale, M.Y., Kim, J., Shaukat, R.A., Noman, M., **Saqib, Q.M.**, Khan, M.U., Dongale, T.D. and Bae, J., 2022. Triboelectric nanogenerator based on biowaste tribopositive delonix regia flowers powder. *Energy Technology*, 10(12), p.2200876 (IF: 3.8).
40. Chougale, M.Y., Khan, M.U., Kim, J., Furqan, C.M., **Saqib, Q.M.**, Shaukat, R.A., Patil, S.R., Mohammad, B., Kwok, H.S. and Bae, J., 2022. Highly Flexible and Asymmetric Hexagonal-Shaped Crystalline Structured Germanium Dioxide-Based Multistate Resistive Switching Memory Device for Data Storage and Neuromorphic Computing. *Advanced Electronic Materials*, 8(10), p.2200332 (IF: 6.2).
41. Chougale, M.Y., Khan, M.U., Kim, J., Shaukat, R.A., **Saqib, Q.M.**, Patil, S.R., Dongale, T.D., Bermak, A., Mohammad, B. and Bae, J., 2022. Bioinspired soft multistate resistive memory device based on silk fibroin gel for neuromorphic computing. *Advanced Engineering Materials*, 24(10), p.2200314 (IF: 3.6).
42. Khan, M.U., Kim, J., Chougale, M.Y., Furqan, C.M., **Saqib, Q.M.**, Shaukat, R.A., Kobayashi, N.P., Mohammad, B., Bae, J. and Kwok, H.S., 2022. Ionic liquid multistate resistive switching characteristics in two terminal soft and flexible discrete channels for neuromorphic computing. *Microsystems & Nanoengineering*, 8(1), p.56 (IF: 8.0).
43. Khan, M.U., Furqan, C.M., Kim, J., Khan, S.A., **Saqib, Q.M.**, Chougale, M.Y., Shaukat, R.A., Kang, M.H., Kobayashi, N.P., Bae, J. and Kwok, H.S., 2022. Asymmetric GaN/ZnO

- engineered resistive memory device for electronic synapses. *ACS Applied Electronic Materials*, 4(1), pp.297-307 (IF: 4.8)
44. Shaukat, R.A., **Saqib, Q.M. (Joint First Author)**, Khan, M.U., Chougale, M.Y. and Bae, J., 2021. Bio-waste sunflower husks powder based recycled triboelectric nanogenerator for energy harvesting. *Energy Reports*, 7, pp.724-731. (IF: 6.631)
 45. Shaukat, R.A., Khan, M.U., **Saqib, Q.M.**, Chougale, M.Y., Kim, J. and Bae, J., 2021. All range highly linear and sensitive humidity sensor based on 2D material TiSi₂ for real-time monitoring. *Sensors and Actuators B: Chemical*, 345, p.130371 (IF: 8.4).
 46. Chougale, M.Y., **Saqib, Q.M. (Joint First Author)**, Khan, M.U., Shaukat, R.A., Kim, J. and Bae, J., 2021. Novel recycled triboelectric nanogenerator based on polymer-coated trash soda can for clean energy harvesting. *Advanced Sustainable Systems*, 5(10), p.2100161. (IF: 7.26).
 47. Khan, M.U., Hassan, G., Shaukat, R.A., **Saqib, Q.M.**, Chougale, M.Y., Kim, J. and Bae, J., 2021. Wide range and highly linear signal processed systematic humidity sensor array using Methylene Blue and Graphene composite. *Scientific Reports*, 11(1), p.16665 (IF: 4.3).
 48. **Saqib, Q.M.**, Khan, M.U., Song, H., Chougale, M.Y., Shaukat, R.A., Kim, J., Bae, J., Choi, M.J., Kim, S.C., Kwon, O. and Bermak, A., 2021. Natural hierarchically structured highly porous tomato peel based tribo-and piezo-electric nanogenerator for efficient energy harvesting. *Advanced Sustainable Systems*, 5(7), p.2100066 (IF: 7.26).
 49. **Saqib, Q.M.**, Shaukat, R.A., Khan, M.U., Chougale, M. and Bae, J., 2020. Biowaste peanut shell powder-based triboelectric nanogenerator for biomechanical energy scavenging and sustainably powering electronic supplies. *ACS Applied Electronic Materials*, 2(12), pp.3953-3963 (IF: 4.8).
 50. Khan, M.U., **Saqib, Q.M.**, Hassan, G. and Bae, J., 2020. All printed organic humidity sensor based on egg albumin. *Sensing and Bio-Sensing Research*, 28, p.100337 (IF: 5.4).
 51. **Saqib, Q.M.**, Khan, M.U. and Bae, J., 2020. Inner egg shell membrane based bio-compatible capacitive and piezoelectric function dominant self-powered pressure sensor array for smart electronic applications. *RSC Advances*, 10(49), pp.29214-29227 (IF: 4.036).
 52. Patil, C.S., Ghode, S.B., Kim, J., Kamble, G., Kundale, S., Mannan, A., Ko, Y., Noman, M., **Saqib, Q.M.**, Patil, S. and Bae, S.Y., 2025. Neuromorphic devices for electronic skin applications. *Materials Horizons* (IF: 14.4). (Featured as Back Cover Article)
 53. Patil, S., Kulkarni G., S, Kim, J., **Saqib, Q.M.**, Noman, M. et al., 2025 Plasma-Assisted Surface Engineering of Binary Metal Chalcogenides: A Path Towards High Energy Efficiency, Electrocatalysts for Water Splitting and Urea Oxidation with Stability Prediction via Machine Learning. *ACS Applied Energy Materials* (IF: 5.5).
 54. CS Patil, M Noman, SB Ghode, SR Patil, J Kim, **QM Saqib, et. al.**, Reactivated Cathode Material from Waste Alkaline Batteries for Supercapacitor Electrodes: Interplay Study Between Pyrolysis Temperature and Surface Morphology, *Small*, 2502556 (2025) (IF: 15.3).
 55. Sawant, Jaydip K., Sourabh B. Ghode... and **Qazi Muhammad Saqib et. al.**, "Human mimetic sensory-interfaced neuromorphic devices and their training mechanism." *International Journal of Extreme Manufacturing* 8, no. 3 (2026): 032008. (IF: 21.3).

	Patent	Saqib, Q.M. and Bae, J., 입자 마찰전기 나노발전장치 및 이를 구비한 웨어러블 전자 장치, 특허청 전기통신심사국 전기심사과 심사관 신호선 신호섭, 발송번호: 9-5-2024-107753308 발송일자: 2024.12.13. (Saqib, Q.M. and Bae, J., Particle triboelectric nanogenerator and wearable electronic devices equipped with them, Shin Hyo-seop, examiner of the Electrical Review Division of the Telecommunications Review Bureau of the Korean Intellectual Property Office, and sending number: 9-5-2024-107753308 Date of shipment: 2024.12.13.)
	Books	- Chougale, M.Y., Shaukat, R.A., Patil, S.R., Noman, M., Kim, J., Saqib, Q.M., Khan, M.U. and Bae, J., Chapter 22: Resistive Switching-based Neuromorphic Devices for Artificial Neural Networks: Functional Materials and Devices, <i>Royal Society of Chemistry</i>, 2023. Hardback ISBN: 978-1-83916-569-6 Qazi Muhammad Saqib, Chapter 8: Polymer Nanocomposites for Resistive Switching Memory in Polymer Nanocomposite Materials: Applications in Integrated Electronic Devices, <i>Wiley</i>, 2021. Print ISBN:9783527347445 Qazi Muhammad Saqib, et al., Polyoxometalate-Based Materials for Sensor Systems, in Polyoxometalate-Based Materials and Applications, <i>Elsevier</i>, pp. 263–292 (2026).
	Selected Conferences	Saqib, Q.M. and Bae, “Lignin derived activated carbon for replacing both tribopositive layer and electrode in triboelectric nanogenerators” In 33RD International Conference on Diamond and Carbon Materials, the Palau de Congressos de Palma, Mallorca, Spain. (10-14 September 2023) Saqib, Q.M. and Bae, “Bio-degradable pressure sensor based on egg shell membrane” In 15th Joint Symposium at Nagasaki University, Japan (2020)
Additional Information		
Funded Research Projects		
	Duration	Jan. 2022 ~ Dec. 31, 2024 (completed)
	Title	Micro-scale Self-powered Energy Storage Systems for Energy Harvesting
	Role	Project Lead (PL)
	Funding Source	National Research Foundation (NRF), South Korea
	Grant Amount	≈154,019 USD
	Description	The goal of the project is to fabricate micro/nano-scale energy harvesting and storage systems from micro/nano-scale particle movements with self-powering features.
	Duration	Jan. 2025 ~ continued
	Title	Green Hydrogen Production from Sea Water
	Role	Participating Researcher Scientist
	Funding Source	National Research Foundation (NRF), South Korea
	Description	Focus to produce hydrogen from sea water.
	Duration	Jan. 2020 ~ Aug. 2027
	Title	Big Data based Ocean Convergence Education AgeNcy (Big OCEAN)
	Role	Participating Researcher
	Funding Source	National Research Foundation (NRF), South Korea
	Description	Focus on big data and metaverse education and research.
	Duration	Mar. 2020 ~ Feb. 2025
	Title	Jeju Sea Grant
	Role	Participating Researcher

Description	Focus on marine researching for energy generation, storage, and sensing.
Student Mentoring/Supervision	- Mentored the following PhD students in their research and thesis: ▫ Mahesh Y. Chougale (PhD): Thesis title – “Advanced materials based self-driven and neuromorphic devices for artificial intelligence systems” (Graduation: 2023) ▫ Rayyan Ali Shaukat (PhD): Thesis title – “Advanced Functional Materials Based Self-powered Nanogenerators and Sensors for Flexible Electronics” (Graduation: 2023) ▫ Swapnil R. Patil (PhD): Thesis title – “Multifaceted Advances for Engineered Materials: Sustainable Energy Harvesting and Electrocatalysis” (Graduation: 2024) ▫ Muhammad Noman (PhD): Thesis title – “Triboelectric and Electrochemical Energy Storage Devices based on Advanced Hybrid Nanomaterials” (Graduation: 2024) ▫ Patil, S. (PhD): Thesis title – “Advanced Nanostructured Materials for Self-Powered and Intelligent Active Sensor Systems.” (Graduation: 2025)
Administrative Responsibilities and Professional Affiliations	- Member of the PhD Thesis Review Committee (2022-present), JNU Jeju, Korea - Advisory Member of the MS Thesis at Dept. of Electronic Engineering, University of Punjab - Registered Engineer of the Pakistan Engineering Council (PEC) - Comprehensive Exam Committee Member, Jeju National University, South Korea ▪ Served as a committee member for MS and PhD comprehensive exams ▪ Provided critical feedback and assessments to ensure academic excellence
Professional Skills and Recognition	
Equipment utilization ability	- Physical Analysis: SEM, TEM, XPS, AFM, EDS, BET, XRD, Contact angle measurement, etc. - Thin Film Deposition: Sputtering, CVD, Ball milling, Solution processing, Vacuum deposition, Volter V-One, Inkjet printing, Screen printing, Metal masking etc. - Optoelectronic Characterizations: Raman spectroscopy, FT-IR, 3D Nano profiling, UV/Vis. Spectrophotometry etc. - Synthesis of the Nanostructured Materials: Hydrothermal, In-situ, Probe sonication, Ball-milling, Heat treatment and processing of materials (tube furnace, muffle furnace) etc. - Surface and Coating Engineering: UV-Ozone Treatment, Plasma treatment etc. - Electrical Measurements (Energy harvesting, Energy storage, Water splitting, sensing, and memory devices): Source measurement unit (SMU), Oscilloscope, Electrochemical impedance spectroscopy, Zeta potential, LCR meter and low noise preamplifier, Dielectric measurements, Measuring C-V, EIS, GCD using potentiostate/Galvanostate, Charge measurements (Keithley) etc. - Simulation platform Handling: COMSOL multi-physics simulation platform etc.
Languages	- English (Excellent), Urdu (Excellent), Arabic (Average), Hindko (Excellent), Korean (Basic)
Talks, Participation in Trainings and Workshops	Keynote Talks: - University of Azad Jammu and Kashmir, Muzaffarabad, Pakistan, Department of Electrical Engineering (2024) - University of Azad Jammu and Kashmir, Muzaffarabad, Pakistan, Department of Electrical Engineering (2022) - National University of Technology (NUTECH), Islamabad, Pakistan (2024) - The Women University Multan, Pakistan (2023)

Other Famous Fellowships and Awards

- University of Poonch, Rawalakot, Pakistan, Department of Electrical Engineering (2022)
- **BK Honor award** (2021), Jeju National University, Korea
- **BK Outstanding Awards** (2020-22), Jeju National University, Korea.
- **Rapid Publication Award** (2022) Jeju National University, Korea
- **BK4 Fellowship Award**-Doctoral Program (2019-2022), Jeju National University, Korea
- **BK21 Fellowship Award**-Master Program (2016-2018), Department of Electronic Engineering, Kyungpook National University, Korea
- **KINGS Scholarship Award** (2016-2018), Department of Electronic Engineering, Kyungpook National University, Korea
- **Best Presentation Award** at 15th Joint Symposium at Nagasaki University Japan (2020)

References

Professor Jinho Bae

Professor, Department of Ocean System Engineering, Jeju National University 102 Jejudaehak-ro, Jeju-si, Jeju Special Self-Governing Province, 63243, Republic of Korea
Director Ocean Information systems Laboratory.
Dean (PD), Big Data based Ocean Convergence Education Agency
Email: baejh@jejunu.ac.kr
Phone: +821085448120

Prof. Seung Goo Lee, Ph.D.

Associate Professor
Department of Chemistry
University of Ulsan, South Korea
Email: lees9@ulsan.ac.kr
Phone: +82-52-712-8007

Prof. Jong-Beom Baek, PhD

Distinguished Professor/Director
Department of Energy and Chemical Engineering
Ulsan National Institute of Science & Technology (UNIST), Ulsan, 44919 South Korea
Phone: +82-52-217-2510
Email: jbbaek@unist.ac.kr